



## Spatial Patterns of Reforestation and Deforestation in Semarang City: From Clustering to Cluster-District Area Comparisons

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### Abstract

**Purpose:** Rapid urbanization in Semarang City has increased land demand, leading to deforestation and the reduction of green spaces, raising environmental concerns such as climate change and biodiversity loss. This research analyzes reforestation and deforestation patterns in Semarang using the Gaussian Mixture Model (GMM) and the Convex Hull-Shoelace Algorithm to define cluster boundaries and calculate cluster areas.

**Methods:** Geospatial data from Indonesia and global forest change data from Earth Engine were used to identify reforestation and deforestation locations. GMM was employed for clustering, and silhouette scores determined the optimal clusters. The Convex Hull-Shoelace method was used to detect cluster boundaries and calculate their areas.

**Results:** Nine clusters were identified for reforestation, mainly in the southern regions, with a silhouette score of 0.51. Three deforestation clusters were found across most of Semarang, except central and northern areas, with a silhouette score of 0.48. The results highlight concentrated reforestation efforts and more dispersed deforestation patterns, indicating significant land-use changes.

**Novelty:** This study applies GMM to reforestation and deforestation in Semarang, which has been underexplored. It enhances understanding of spatial patterns by determining optimal cluster numbers and visualizing them with Convex Hull-Shoelace Algorithm, contributing to improved forest management insights.

**Keywords:** Geospatial Clustering, Forest Cover Change, Semarang City, Gaussian Mixture Model, Convex Hull

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### INTRODUCTION

The increasing population in urban Semarang City has driven deforestation and the reduction of green spaces, fueled by rapid urbanization and development, raising concerns about environmental impacts such as climate change, air quality deterioration, and biodiversity loss. Between 1990 and 2015, the built-up area in Metropolitan Semarang increased from 74.85 km<sup>2</sup> to 292.74 km<sup>2</sup>, resulting in forest conversion for residential and industrial purposes [1]. This expansion has led to a decrease in green vegetation cover and an increase in built-up density, as indicated by negative correlations between NDBI (Normalized Difference Built-up Index) and NDVI (Normalized Difference Vegetation Index) [1]. To address these issues, the Semarang City Government implemented regulations to manage trees in public green spaces, including prohibitions on illegal logging [2]. However, challenges persist, such as land disputes [3] and construction in flood-prone areas due to population growth and high land prices [4]. These factors contribute to ongoing deforestation and environmental degradation in Semarang City.

Clustering is a technique used to group data points based on their similarities, where points in the same cluster are more similar to each other than to those in other clusters [5]. In spatial data, clustering helps reveal patterns and regions of interest, such as areas with similar land use, vegetation, or population densities, making it valuable for applications like urban planning and environmental management [6]. State-of-the-art clustering methods include algorithms like K-Means, DBSCAN, and Gaussian Mixture Models (GMM) [6], [7]. K-Means is simple and efficient but struggles with non-spherical clusters [8],

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while DBSCAN excels at identifying clusters of arbitrary shapes and handling noise but can be sensitive to parameter tuning [9]. GMM, however, offers a significant advantage through its probabilistic approach, allowing points to belong to multiple clusters with varying degrees of membership [10]. This flexibility makes GMM more adept at modeling complex, overlapping data distributions and non-spherical clusters, providing a deeper understanding of spatial patterns compared to traditional clustering methods [11]. Additionally, GMM also demonstrated strong performance in the Silhouette Index for determining the optimal number of clusters in the data [12].

Spatial clustering is a process of grouping geospatial data points based on their similarities to identify patterns or regions of interest [13]. GMM has been applied to various clustering and classification tasks in remote sensing [14] and geospatial analysis [15], [16]. In Indonesia, GMM was used to cluster villages in West Kalimantan based on development indices, revealing distinct patterns across social, economic, and environmental factors [15]. GMM has also been employed in image processing to assess river cleanliness by detecting and quantifying debris [16]. Research on spatial clustering of deforestation has been widely conducted, but the methods used still predominantly rely on K-Means [17], [18], which may be limited in capturing complex and overlapping spatial patterns that could be better addressed by more flexible models like GMM.

A convex hull is the smallest convex shape that completely encloses a set of points, forming the outer boundary, and is used in computational geometry to define the spatial extent of data clusters [19], [20]. Convex hull algorithms are widely applied in image processing to extracting image components [21] and shape analysis to efficiently grouping points [22]. Convex hull can be applied to boundary detection [23] in spatial clustering [24], and when used as a boundary detection algorithm for data points clustered by GMM, it helps visualize the spatial extent of each cluster and estimate its area for further spatial analysis. The Shoelace algorithm can be applied to calculate the area of a convex polygon [25], where the area formed by the convex hull on the clusters can be computed. This area estimation enhances spatial analysis by providing quantifiable metrics related to the extent of reforestation and deforestation clusters.

Reforestation and deforestation significantly impact environmental sustainability and land use in urban areas. A few researchers have focused on clustering methods to analyze these spatial phenomena, particularly utilizing Gaussian Mixture Models (GMM) and the Silhouette Index. However, there have been limited studies concerned with the application of GMM in the context of reforestation and deforestation in Semarang. Therefore, this research intends to explore the clustering of point data related to reforestation and deforestation activities in Semarang, determining the number of clusters using the Silhouette Index. The objectives of this research are to detect boundaries of the identified clusters, calculate their areas using the Convex Hull-Shoelace method, and provide a spatial analysis of changes in forest areas in Semarang.

## **METHODS**

In this section, the methods used will be discussed. This research is divided into several stages. First, datasets containing regional boundaries of Semarang City, along with the locations of reforestation and deforestation points, were collected. Next, clustering was performed using the GMM algorithm. After that, the optimal clusters were determined through grid search based on silhouette score evaluation. Finally, the cluster boundaries were created, and their area was calculated using the Convex Hull method and Shoelace Algorithm.

### **Dataset**

This study used two datasets. The first dataset is the regional boundary dataset in Indonesia, which contains boundaries ranging from rural areas to provinces in Indonesia. This dataset is sourced from Indonesia Geospatial [26]. This data is utilized to limit the acquisition of the second dataset and for visualizing regional boundaries. In this research, the Indonesian regional boundary dataset was placed on Kaggle [27] and accessed in Google Colab through the Kaggle API due to the dataset's large size. The first dataset is restricted to only cover the scope of Semarang City, as shown in Figure 1.

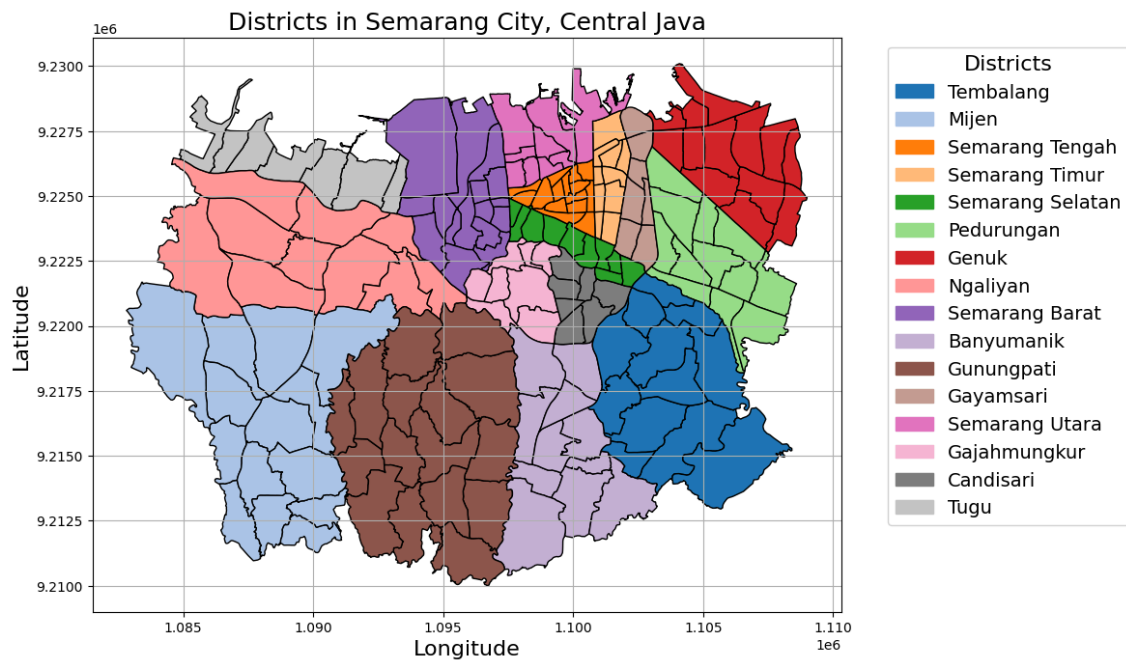


Figure 1. Regional boundaries in Semarang City

The second dataset is a global forest change dataset [28], which can be accessed through the Earth Engine (EE) API. This dataset contains information on global forest change from year to year, covering the period from 2000 to 2012 [28]. The ‘gain’ and ‘loss’ layers, representing reforestation and deforestation that occurred during this period, were extracted [29]. The extraction of reforestation and deforestation point locations from the dataset was performed by utilizing these layers and filtering pixels that reflect either reforestation or deforestation, subsequently converting them into vector points representing the center of each pixel. This process enables cluster analysis to identify reforestation distribution patterns in the studied area. The process was limited to the Semarang City area with a resolution of one meter per pixel, resulting in more detailed and relevant data for research purposes. Table 1 shows the total data points that were successfully extracted. Figures 2 and 3 illustrate the locations of reforestation and deforestation points in Semarang City, respectively.

Table 1. Number of Reforestation and Deforestation Location Points

| Target        | Total |
|---------------|-------|
| Reforestation | 111   |
| Deforestation | 1530  |

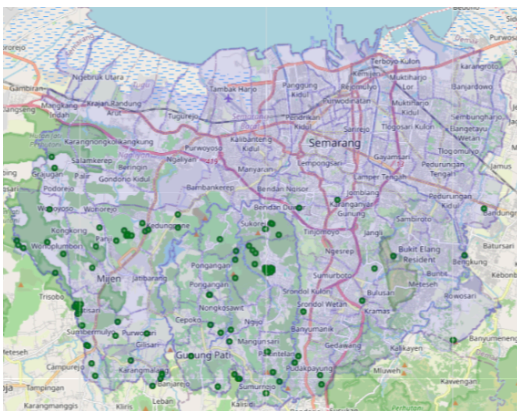


Figure 2. Reforestation Points in Semarang City

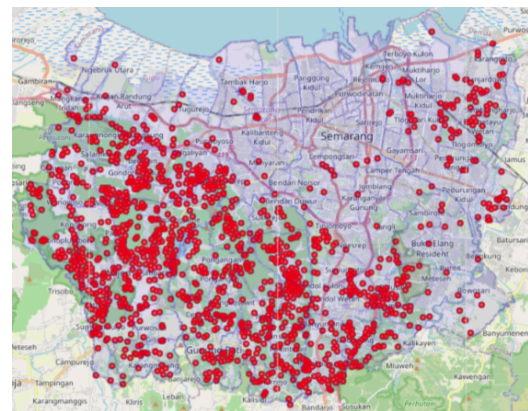


Figure 3. Deforestation Points in Semarang City

### Clustering Model

The GMM model was applied to cluster the reforestation and deforestation points in Semarang City. The algorithm process in GMM includes both the model and parameter estimation process. The estimation process is performed automatically to adjust the optimal parameters in the GMM.

- 1) **The Gaussian Mixture Model** can be expressed as a linear combination of several Gaussian distributions.

$$P(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k) \quad (1)$$

Where  $P(X)$  is the probability distribution of data  $X$ .  $K$  is the number of cluster components in the model.  $\pi_k$  is the weight (prior) of the  $k$ -th component, with the condition that  $\sum_{k=1}^K \pi_k = 1$  and  $\pi_k \geq 0$ . And  $\mathcal{N}(X|\mu_k, \Sigma_k)$  is the Gaussian probability distribution function defined as.

$$\mathcal{N}(X|\mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} \exp\left(-\frac{1}{2}(X - \mu_k)^T \Sigma_k^{-1} (X - \mu_k)\right) \quad (2)$$

Where  $\mu_k$  is the mean of the  $k$ -th component.  $\Sigma_k$  is the covariance matrix of the  $k$ -th component.  $d$  is the dimension of the data  $X$ . And  $|\Sigma_k|$  is the determinant of the covariance matrix  $\Sigma_k$ .

- 2) Then, **the parameter estimation process** is conducted to determine the model parameters  $\pi_k$  (proportion),  $\mu_k$  (mean), and  $\Sigma_k$  (covariance) in GMM using the Expectation-Maximization (EM) algorithm.. The EM process consists of two steps: the Expectation step (E-step) and the Maximization step (M-step).

**E-step** equation calculates the posterior probability that each data point  $X_i$  originates from the  $k$ -th component:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(X_i|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(X_i|\mu_j, \Sigma_j)} \quad (3)$$

Where  $\gamma_{ik}$  is the probability that data  $X_i$  originates from the  $k$ -th component. Then, the **M-step** updates the model parameters based on the results from the E-step.

$$\pi_k = \frac{N_k}{N} \quad (4)$$

$$\mu_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} X_i \quad (5)$$

$$\Sigma_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} (X_i - \mu_k)(X_i - \mu_k)^T \quad (6)$$

Where  $N$  is the total number of data points.  $\Sigma_k = \sum_{i=1}^N \gamma_{ik}$  is the number of data points assigned to the  $k$ -th cluster.

### Finding Optimal Cluster

During this process, a **grid search** was conducted to find the optimal number of clusters, as it allowed the determination of the most suitable number of clusters for the dataset used. The range of clusters tested on the reforestation and deforestation data was between two and ten clusters to reduce the complexity of clustering in both the analysis and visualization. By testing various values for the number of clusters, the grid search helped identify the configuration that provided the best separation between the data, resulting

in more representative clusters. This process focused on finding the optimal parameters for the GMM model, which were later evaluated using silhouette scores to assess the quality of the resulting clusters.

**Silhouette Score** is an evaluation method that measures how well data points are clustered into the correct groups [30]. For each data point  $i$ , the silhouette value is calculated:

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (7)$$

Where  $a(i)$  is the average distance between point  $i$  and all other points in the same cluster (intra-cluster distance).  $b(i)$  is the average distance between point  $i$  and the points in the nearest different cluster (inter-cluster distance). The function  $\max(a(i), b(i))$  is used to normalize the value between the intra-cluster distance and the distance to the nearest cluster. The silhouette score ranges from -1 to 1. If it approaches 1, the data points are correctly clustered, and if it approaches -1, the data points are incorrectly clustered.

#### Cluster Boundaries and Area Calculation

Next, the boundaries of the clusters were detected and their areas calculated using the Convex Hull method. By applying the **Convex Hull algorithm**, the outermost boundary of each cluster was determined, resulting in an area that encompasses all points identified within the cluster. At this stage, the Convex Hull algorithm plays a key role in efficiently grouping scattered points, similar to drawing a string around them. A commonly used approach for constructing a convex hull starts by selecting the lowest point as a reference. The remaining points are then sorted based on their polar angle relative to this reference. After establishing the order, the algorithm proceeds to "wrap" the points, checking if each subsequent point maintains the convex shape. Any points that cause an "inward bend" are discarded, leaving only the outermost points that form the boundary enclosing the entire set. Figure 4 shows an example of the input and output from the use of the Convex Hull in a cluster.

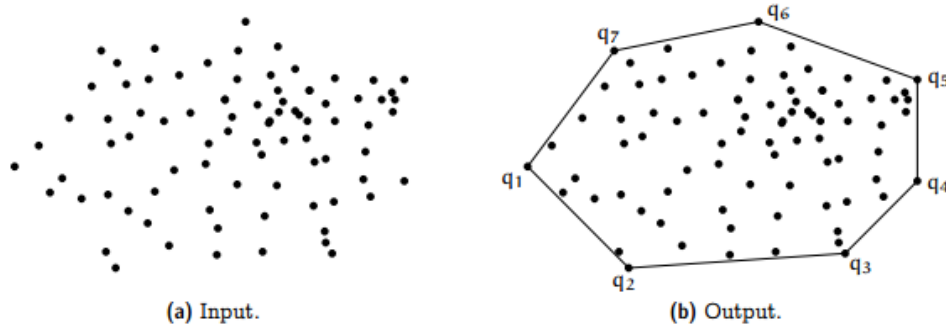


Figure 4. Example of a Convex Hull used to determine the area of the cluster

For each polygon formed by the Convex Hull in each cluster, the area will be calculated using the **Shoelace Algorithm**. For each outermost point of the cluster that defines the polygon's boundary, the area of the polygon ( $A$ ) will be computed using the following formula:

$$A = \frac{1}{2} \left| \sum_{i=1}^n (x_i \cdot y_{i+1} - y_i \cdot x_{i+1}) \right| \quad (8)$$

Where  $n$  is the number of vertices in the polygon, and the last point  $(x_n, y_n)$  is connected back to the first point  $(x_1, y_1)$ .

#### RESULT AND DISCUSSION

In this section, the research results will be discussed and reviewed. To apply this method, Google Colab was used at the free user level, which provides 12.7 GB of RAM and 107.7 GB of storage, utilizing the Python programming language. After applying the methods described earlier, the research findings were obtained. First, regional information for Semarang City will be presented, followed by graphical representations of the optimal clusters for both reforestation and deforestation. Lastly, more detailed information on the clusters formed by the clustering model will be provided.

### Regional Information in Semarang City

In this study, the area of Semarang City was calculated to be 372.97 km<sup>2</sup> based on the total area calculation from the Indonesian regional boundary dataset. Additionally, the area and the total number of reforestation and deforestation points for each district in Semarang City were calculated. This information is presented in Table 2. Regarding the area and distribution of locations in Semarang City, most of the reforestation and deforestation locations are concentrated in the western to southeastern regions. The districts with the highest concentrations of reforestation and deforestation, respectively, are Mijen District, Gunungpati District, Ngaliyan District, Banyumanik District, and Tembalang District. These five districts are also the largest in Semarang City, indicating that the size of the area may influence the distribution of reforestation and deforestation locations, as larger areas are generally associated with more intense vegetation activity. On the other hand, the central areas of Semarang City, such as the central and northern parts, show fewer distribution points, reflecting lower vegetation activity. Therefore, further research is needed to understand other factors that may influence the distribution of reforestation and deforestation locations.

Table 2. Information on District Areas and Distribution of Reforestation/Deforestation Points in Semarang City

| Districts                             | Area Size (km <sup>2</sup> ) | Ratio of District Area to the Area of Semarang City (%) | Total Location Points |               |
|---------------------------------------|------------------------------|---------------------------------------------------------|-----------------------|---------------|
|                                       |                              |                                                         | Reforestation         | Deforestation |
| Gunungpati                            | <b>60,25</b>                 | <b>16,15</b>                                            | 42                    | 426           |
| Mijen                                 | 57,98                        | 15,54                                                   | <b>53</b>             | <b>428</b>    |
| Ngaliyan                              | 45                           | 12,07                                                   | 2                     | 278           |
| Tembalang                             | 41,23                        | 11,05                                                   | 4                     | 80            |
| Banyumanik                            | 32,15                        | 8,62                                                    | 6                     | 180           |
| Semarang Barat<br>(West Semarang)     | 25,36                        | 6,80                                                    | -                     | 11            |
| Genuk                                 | 22,69                        | 6,08                                                    | -                     | 51            |
| Pedurungan                            | 21,98                        | 5,89                                                    | 1                     | 46            |
| Tugu                                  | 14,73                        | 3,95                                                    | -                     | 9             |
| Semarang Utara<br>(North Semarang)    | 11,44                        | 3,07                                                    | -                     |               |
| Gajahmungkur                          | 9,71                         | 2,60                                                    | 1                     | 7             |
| Candisari                             | 6,69                         | 1,79                                                    | 1                     | 2             |
| Gayamsari                             | 6,52                         | 1,75                                                    | -                     | 6             |
| Semarang Selatan<br>(South Semarang)  | 6,22                         | 1,67                                                    | -                     | 3             |
| Semarang Timur<br>(East Semarang)     | 5,64                         | 1,51                                                    | -                     | 2             |
| Semarang Tengah<br>(Central Semarang) | 5,38                         | 1,44                                                    | -                     | -             |
| Total                                 | 372,97                       | 100                                                     | 111                   | 1530          |

### Optimal Clusters

In this study, the optimal number of clusters for both models, reforestation and deforestation, has been determined. Based on the analysis results during the data processing stage, the reforestation model indicates that the most optimal number of clusters is 9, with a silhouette score of 0.51, as shown in Figure 5. Meanwhile, for the deforestation model, the optimal number of clusters obtained is 3, with a silhouette score of 0.48, as displayed in Figure 6.

The findings related to the optimal clusters reveal significant differences in the spatial distribution complexity between reforestation and deforestation. The number of locations for reforestation is significantly higher compared to deforestation, resulting in more clusters with relatively smaller areas. This suggests that reforestation activities in Semarang City tend to be concentrated in smaller, specific areas with similar characteristics, leading to the formation of more but smaller-sized clusters. In contrast, deforestation exhibits a different pattern, where fewer clusters are formed, but each encompasses a much larger area. This reflects that deforestation may occur in broader and less organized regions, leading to the formation of larger clusters.

This difference suggests that reforestation is conducted with a more planned and targeted approach, whereas deforestation is influenced by factors such as forest management policies, human activities, or local economic conditions that can lead to deforestation over larger areas. Additionally, the use of the

highest silhouette score in this analysis has assisted in determining the optimal clusters, providing a clearer understanding of the structure and distribution patterns of both activities.

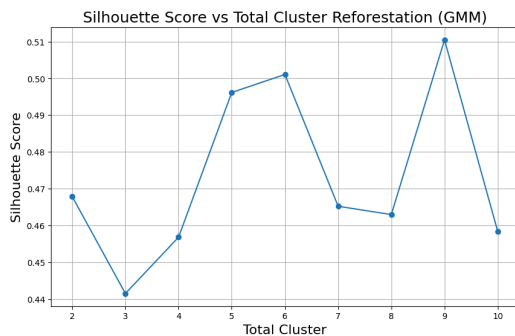


Figure 5. Grid search for optimal cluster search in the reforestation data model

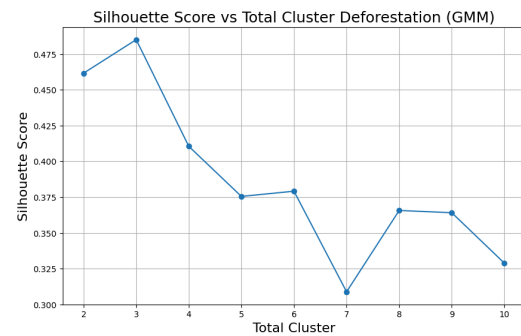


Figure 6. Grid search for optimal cluster search in the deforestation data model

### Detail Information on Clusters

The total number of points in each cluster, the estimated area of each cluster using the Convex Hull method, and the percentage of districts covered by each cluster for both reforestation and deforestation data in Semarang City have been calculated. This information is presented in Table 3. Additionally, visualizations are provided in Figures 6 and 7, which display the cluster areas for reforestation and deforestation data in Semarang City, respectively.

For the reforestation data, Cluster 2, which contains the highest number of reforestation points, surprisingly has a smaller estimated area compared to Cluster 3, which has the largest estimated area despite having fewer reforestation points. This suggests that reforestation in Cluster 2 is likely concentrated in denser areas, while in Cluster 3, reforestation is more spread out over a larger area but with lower density. For the deforestation data, the pattern observed shows that the cluster with the largest area also contains the highest number of deforestation points, while smaller clusters exhibit a consistent relationship between the size of the deforested area and the number of points in each cluster. This indicates that deforestation may occur over larger areas with more evenly distributed activity compared to reforestation.

In the reforestation data, nine clusters were formed, primarily located in the southern part of Semarang City, particularly in Mijen and Gunungpati districts. In contrast, the deforestation clusters tend to cover most parts of Semarang City, except for the central and northern regions, such as Semarang Tengah and Semarang Utara districts, indicating that deforestation is more concentrated in areas with easier accessibility or more open land. Additionally, it can be observed that the northwestern and northeastern regions tend to experience deforestation, while no reforestation occurs in these areas.

This difference suggests that the distribution of deforestation may be influenced by factors such as accessibility and land use. Further research is needed to verify the patterns formed in these reforestation and deforestation clusters and to explore the factors that may affect their distribution with additional and updated data. This study is expected to serve as a reference for forestry policies in Semarang City.

Table 3. Cluster Information on Reforestation and Deforestation Data

| Cluster       | Total Location Points | Estimated Area (km <sup>2</sup> ) | Percentage of Districts per Cluster                |
|---------------|-----------------------|-----------------------------------|----------------------------------------------------|
| Reforestation |                       |                                   |                                                    |
| 1             | 15                    | 3,51                              | Mijen (100%)                                       |
| 2             | 26                    | 10,17                             | Gunungpati (96,15%); Gajahmungkur (3,85%)          |
| 3             | 23                    | 23,52                             | Gunungpati (78,26%); Banyumanik (21,74%)           |
| 4             | 2                     | -                                 | Tembalang (50%); Pedurungan (50%)                  |
| 5             | 10                    | 2,3                               | Mijen (100%)                                       |
| 6             | 21                    | 10,13                             | Mijen (100%)                                       |
| 7             | 2                     | -                                 | Ngaliyan (100%)                                    |
| 8             | 5                     | 13,06                             | Tembalang (60%); Banyumanik (20%); Candisari (20%) |
| 9             | 7                     | 4,12                              | Mijen (100%)                                       |
| Deforestation |                       |                                   |                                                    |



|   |     |        |                                                                                                                                 |
|---|-----|--------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 | 610 | 123.76 | Gunungpati (56,65%); Banyumanik (29,56%); Tembalang (12,15%); Gajahmungkur (0,99%); Candisari (0,33%); Semarang Selatan (0,33%) |
| 2 | 808 | 155.77 | Genuk (45,54%); Pedurungan (41,07%); Tembalang (5,36%); Gayamsari (5,36%); Semarang Timur (1,79%); Semarang Selatan (0,89%)     |
| 3 | 112 | 52.96  | Mijen (52,97%); Ngaliyan (34,41%); Gunungpati (10,02%); Semarang Barat (1,36%); Tugu (1,11%); Gajahmungkur (0,12%)              |

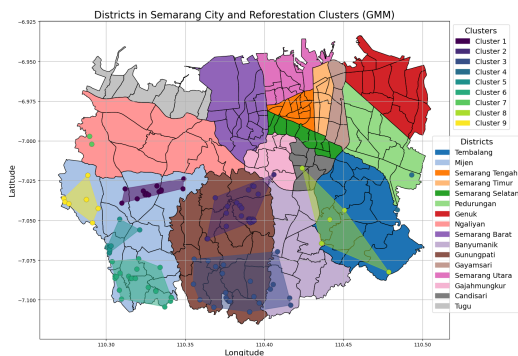


Figure 6. Visualization of cluster locations in the reforestation data in Semarang City

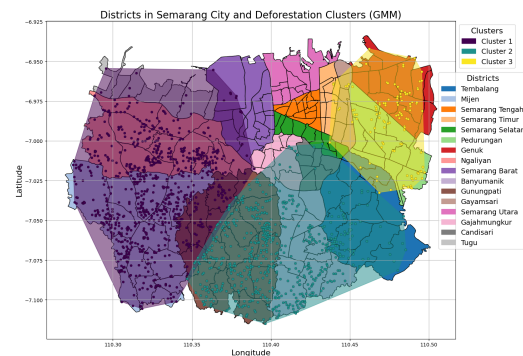


Figure 7. Visualization of cluster locations in the deforestation data in Semarang City

## CONCLUSION

This study successfully identified the distribution patterns of reforestation and deforestation in Semarang City using the Gaussian Mixture Model (GMM), detected cluster boundaries through the Convex Hull method, and applied the Shoelace Algorithm to calculate their areas. The findings revealed that reforestation has fewer location points compared to deforestation, resulting in nine clusters for reforestation and three clusters for deforestation. Additionally, the districts in Semarang City impacted by these activities were identified. These results suggest that reforestation tends to follow a more structured and planned approach, while deforestation is likely influenced by factors such as land-use policies or accessibility. However, this research is limited by the use of global forest change data from 2000 to 2012, which does not account for more recent developments. Future research is recommended to update the dataset and include other factors, such as changes in forestry policies and local economic activities. This study provides valuable insights into forest area changes and offers important contributions to decision-making related to forest management in Semarang City.

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